

Kotlin String Manipulation

Complete grouped reference with examples for Android/Kotlin interviews

Key idea: Kotlin String is immutable. Most operations return a new value. Use CharArray when you need indexed mutation and StringBuilder for repeated building/modification.

Complete Grouping Map

```
KOTLIN STRING MANIPULATION
|
+-- 1. Creation & Basic Properties
|   +-- String literals / multiline strings
|   +-- length
|   +-- isEmpty()
|   +-- isBlank()
|   +-- String templates
|
+-- 2. Character Access
|   +-- text[index]
|   +-- get()
|   +-- getOrNull()
|   +-- first() / last()
|   +-- firstOrNull() / lastOrNull()
|
+-- 3. Searching & Indexes
|   +-- contains()
|   +-- startsWith() / endsWith()
|   +-- indexOf() / lastIndexOf()
|   +-- indexOfFirst() / indexOfLast()
|   +-- find()
|
+-- 4. Comparison
|   +-- == / !=
|   +-- ===
|   +-- equals(ignoreCase = true)
|   +-- compareTo()
|
+-- 5. Extracting Parts
|   +-- substring()
|   +-- substringBefore() / substringAfter()
|   +-- substringBeforeLast() / substringAfterLast()
|   +-- take() / takeLast()
|   +-- drop() / dropLast()
|
+-- 6. Removing & Trimming
|   +-- removePrefix() / removeSuffix()
|   +-- removeRange()
|   +-- trim() / trimStart() / trimEnd()
|   +-- filterNot()
|
+-- 7. Replacing & Transforming
|   +-- replace() / replaceFirst()
|   +-- uppercase() / lowercase()
|   +-- replaceFirstChar()
|   +-- reversed()
|
+-- 8. Splitting & Joining
|   +-- split()
|   +-- lines()
|   +-- chunked()
|   +-- joinToString()
|
+-- 9. Character Testing
|   +-- isLetter() / isDigit()
|   +-- isLetterOrDigit()
|   +-- isWhitespace()
|   +-- isUpperCase() / isLowerCase()
```

```

+-- 10. Collection-style Operations
|   +-- filter() / filterNot()
|   +-- map() / mapIndexed()
|   +-- any() / all() / none()
|   +-- count()
|   +-- find() / firstOrNull()
|   +-- groupingBy() / eachCount()
|
+-- 11. Conversions
|   +-- toCharArray()
|   +-- toList() / toSet()
|   +-- toInt() / toIntOrNull()
|   +-- toLong() / toLongOrNull()
|   +-- toString()
|
+-- 12. Mutable String Processing
|   +-- StringBuilder
|       +-- append() / insert()
|       +-- delete() / replace()
|       +-- reverse() / toString()
|
+-- 13. Regex
|   +-- Regex()
|   +-- matches() / containsMatchIn()
|   +-- find() / findAll()
|   +-- replace() / split()
|
+-- 14. Interview Algorithms
    +-- Reverse String
    +-- Palindrome
    +-- Character Frequency
    +-- First Non-Repeating Character
    +-- Duplicate Characters
    +-- Remove Duplicates
    +-- Anagram
    +-- Reverse Words / Reverse Each Word
    +-- Count vowels & consonants
    +-- String Rotation
    +-- Longest Common Prefix
    +-- Longest Word
    +-- Character Sorting

```

1. Creation & Basic Properties

```
val name = "Rakesh"
val text = "Kotlin"
val multiline = """
    Android
    Kotlin
    Compose
    """.trimIndent()

println(text.length)           // 6
println(text.isEmpty())       // false
println("   ".isBlank())      // true
println("Hello $name")
println("Length = ${text.length}")
```

2. Character Access

```
val text = "Kotlin"

println(text[0])               // K
println(text.get(1))           // o
println(text.getOrNull(100))   // null
println(text.first())          // K
println(text.last())           // n
println(text.firstOrNull())
println(text.lastOrNull())
```

3. Searching & Indexes

Remember: there is no `firstIndexOf()`. `indexOf()` returns the first occurrence of a value. `indexOfFirst { }` returns the first index matching a condition.

```
val text = "banana"

println(text.contains("ana"))   // true
println(text.startsWith("ban")) // true
println(text.endsWith("ana"))   // true

println(text.indexOf('a'))      // 1
println(text.lastIndexOf('a'))  // 5
println(text.indexOf('z'))      // -1

println(text.indexOfFirst { it == 'n' }) // 2
println(text.indexOfLast { it == 'n' })  // 4

val mixed = "abc123"
println(mixed.indexOfFirst { it.isDigit() }) // 3
println(mixed.find { it.isDigit() })         // '1'
```

4. Comparison

```
val a = "Kotlin"
val b = "Kotlin"

println(a == b)           // structural equality
println(a != b)
println(a === b)          // referential identity
println("Kotlin".equals("KOTLIN", ignoreCase = true))
println("abc".compareTo("abd"))
```

5. Extracting Parts

```
val email = "rakesh@gmail.com"

println(email.substring(0, 6))      // rakesh
println(email.substringBefore("@")) // rakesh
println(email.substringAfter("@"))  // gmail.com

val path = "users/profile/photo.jpg"
```

```
println(path.substringBeforeLast("/"))
println(path.substringAfterLast("/"))

val text = "Android"
println(text.take(3))           // And
println(text.takeLast(3))      // oid
println(text.drop(3))          // roid
println(text.dropLast(3))      // Andr
```

6. Removing & Trimming

```
val text = "Android Developer"

println(text.removePrefix("Android "))
println(text.removeSuffix(" Developer"))
println("Android".removeRange(0, 3))

val input = "  Rakesh  "
println(input.trim())
println(input.trimStart())
println(input.trimEnd())

println("Rak123esh".filterNot { it.isDigit() }) // Rakesh
```

7. Replacing & Transforming

```
println("I love Java".replace("Java", "Kotlin"))
println("banana".replace('a', 'o'))
println("Java Java".replaceFirst("Java", "Kotlin"))
println("android".uppercase())
println("ANDROID".lowercase())
println("kotlin".replaceFirstChar { it.uppercase() })
println("Android".reversed()) // diordnA
```

8. Splitting & Joining

```
val csv = "Kotlin,Java,Python"
val items = csv.split(",")
println(items)

println(items.joinToString(" | "))

println("123456789".chunked(3))
// [123, 456, 789]

val multi = "Kotlin\nJava\nCompose"
println(multi.lines())
```

9. Character Testing

```
val ch = 'A'

println(ch.isLetter())
println(ch.isDigit())
println(ch.isLetterOrDigit())
println(ch.isWhitespace())
println(ch.isUpperCase())
println(ch.isLowerCase())
```

10. Collection-style Operations

```
val text = "abc123"

println(text.filter { it.isDigit() }) // 123
println(text.filterNot { it.isDigit() }) // abc
println(text.map { it.uppercaseChar() })
println(text.any { it.isDigit() }) // true
println(text.all { it.isLetterOrDigit() }) // true
println(text.none { it.isWhitespace() }) // true
println(text.count { it.isDigit() }) // 3
```

```
println(text.firstOrNull { it.isDigit() }) // 1

text.mapIndexed { index, char ->
    println("$index -> $char")
}

println("banana".groupingBy { it }.eachCount())
// {b=1, a=3, n=2}
```

11. Conversions

```
val text = "Kotlin"

val chars = text.toCharArray()
val list = text.toList()
val set = text.toSet()

println(String(chars))
println("123".toInt())
println("ABC".toIntOrNull()) // null
println("123".toLongOrNull())
println(100.toString())
```

12. StringBuilder

```
val sb = StringBuilder("Kotlin")

sb.append(" Android")
sb.insert(0, "Learn ")
sb.delete(0, 6)
sb.replace(0, 6, "Java")
sb.reverse()

println(sb.toString())

// Useful when repeatedly constructing/modifying text.
```

13. Regex

```
val text = "Order123 costs 500 rupees"
val regex = Regex("\\d+")

println(regex.containsMatchIn(text))
println(regex.find(text)?.value)

val numbers = regex.findAll(text)
    .map { it.value }
    .toList()

println(numbers) // [123, 500]

println("Rakesh      Android"
    .replace(Regex("\\s+"), " "))
```

14. Interview Algorithms

```
// Reverse
fun reverse(s: String) = s.reversed()

// Palindrome
fun isPalindrome(s: String): Boolean =
    s == s.reversed()

// Character frequency
fun frequency(s: String) =
    s.groupingBy { it }.eachCount()

// First non-repeating
fun firstUnique(s: String): Char? {
    val freq = s.groupingBy { it }.eachCount()
```

```

        return s.firstOrNull { freq[it] == 1 }
    }

    // Anagram
    fun isAnagram(a: String, b: String): Boolean =
        a.length == b.length &&
        a.groupingBy { it }.eachCount() ==
        b.groupingBy { it }.eachCount()

    // Remove duplicates while preserving order
    fun removeDuplicates(s: String): String {
        val seen = mutableSetOf<Char>()
        return s.filter { seen.add(it) }
    }

    // Reverse words
    fun reverseWords(s: String): String =
        s.split(" ").reversed().joinToString(" ")

    // String rotation
    fun isRotation(a: String, b: String): Boolean =
        a.length == b.length && (a + a).contains(b)

    // Longest word
    fun longestWord(s: String): String? =
        s.split(" ").maxByOrNull { it.length }

    // Sort characters
    fun sortChars(s: String): String =
        s.toCharArray().sorted().joinToString("")

```

What to Prioritize for Interviews

Know the API groups, but focus especially on `indexOf/indexOfFirst`, substring extraction, `filter/map/groupingBy`, String immutability, `CharArray`, `StringBuilder`, Regex basics, and solving palindrome/anagram/frequency/unique-character problems while explaining time and space complexity.